

Mechanism of stepwise electron transfer in six-transmembrane epithelial antigen of the prostate (STEAP) 1 and 2

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Abstract

Six transmembrane epithelial antigen of the prostate (STEAP) is a family of membrane-embedded hemoproteins with four members, STEAP1-4, all of which have a transmembrane domain (TMD) that chelates a heme prosthetic group. STEAP2-4, but not STEAP1, have an intracellular oxidoreductase domain (OxRD) so that an electron transfer chain composed of NADPH, FAD, and heme is established to mediate electron transfer across cell membranes. However, it is not known whether STEAP1 can establish a physiologically relevant electron transfer chain. Here we show that reduced FAD binds to STEAP1 and enables reduction of the heme. We also show that a soluble cytochrome *b*₅ reductase can dock on STEAP1 and serve as a surrogate OxRD to reduce the heme. These results provide the first evidence that STEAP1 can support a cross-membrane electron transfer chain. It is not clear whether FAD, which relays electrons from NADPH to heme and interacts with both OxRD and TMD, remains constantly bound to the STEAPs. We found that FAD reduced by STEAP2 can be utilized by STEAP1, supporting the hypothesis that FAD is diffusible rather than staying bound to STEAP2. We determined the structure of human STEAP2 in complex with NADP⁺ and FAD to an overall resolution of 3.2 Å by cryo-electron microscopy. The structure of STEAP2 shows that the two cofactors bind similarly to those in the STEAP4 structure and thus a diffusible FAD is likely a general feature of the electron transfer mechanism in the STEAPs. The structure of STEAP2 also shows that its extracellular regions are less structured than those of STEAP4 or STEAP1, and further experiments show that STEAP2 reduces Fe³⁺-NTA with a rate significantly slower than STEAP1. These results establish a solid foundation for understanding the function and mechanisms of STEAP family of proteins.

eLife assessment

This study provides **useful** insights into the mechanisms of electron transport in STEAP proteins, consistent with current models. The work strengthens and supports previously published biochemical and structural data, and the experimental results are of **solid** technical quality. The manuscript will be of interest to colleagues who work on STEAP proteins and related electron transfer systems.

Introduction

Six transmembrane epithelial antigen of the prostate 1 (STEAP1) was first discovered owing to its high level of expression in prostate cancer cells.¹ Discovery of STEAP2-4 soon followed, and further analyses show that STEAP2-4 have metal ion reductase activities.^{2, 3} STEAP3 was identified as a ferrireductase required for iron uptake in erythroid cells.^{3, 4} STEAP2-4 are also found overexpressed in many types of cancer cells, suggesting their involvement in cancer initiation or progression.^{1, 5, 6}

Each STEAP protein has a transmembrane domain (TMD) which consists of six transmembrane helices (TM), and STEAP2-4, but not STEAP1, also have a N-terminal intracellular oxidoreductase domain (OxRD).^{7, 8} TMD of the STEAP family of proteins are homologous to membrane-embedded reductases including mammalian NADPH oxidases (NOX) and dual oxidases (DUOX), and bacterial and yeast ferric reductases (FRE).⁹ The structures of NOX and DUOX show that their TMD binds two heme prosthetic groups, each ligated to a pair of conserved histidine residues from TMs 3 and 5; one heme is close to the intracellular side and the other close to the extracellular side of the TMD.^{10–14} In NOX or DUOX, the OxRD binds both NADPH and FAD, and the cross-membrane electron transfer chain starts with hydride transfer from NADPH to FAD, and then sequentially to the intracellular and extracellular hemes, and finally to the substrate. The structure of STEAP4, on the other hand, shows that only one heme is present in the TMD and it corresponds to the extracellular heme in NOX or DUOX.¹⁵ In STEAP4, the FAD straddles OxRD and TMD with the isoalloxazine ring of FAD binding at the equivalent position of the intracellular heme in NOX and DUOX and the nucleotide moiety of FAD binding to the OxRD (Fig. S1). While this configuration allows electron transfer between FAD and heme, the isoalloxazine ring is too far away from the nicotinamide ring of NADPH to receive hydride. Thus, the isoalloxazine ring of FAD must dissociate from the TMD and move closer to the nicotinamide ring of NADPH for hydride transfer. Although the isoalloxazine ring of FAD must assume different conformations during the redox cycles in STEAPs, its adenosine moiety could stay bound to the OxRD. However, here we show evidence that FAD does not stay tightly bound to the STEAP protein and can become diffusible after its reduction.

STEAP1 has been considered not be able to establish an electron transfer chain due to its lack of an OxRD, however, recent structures of STEAP1 and STEAP4 offer some hints otherwise.^{15, 16} In the STEAP4 structure, the isoalloxazine ring of FAD is coordinated by residues in TMD that are conserved in STEAP1. The structure of STEAP1 aligns well with that the TMD of STEAP4, and although FAD is not present in the STEAP1 structure,¹⁶ we showed previously that purified STEAP1 contains residual FAD, and that FAD binds to STEAP1 with K_D of $\sim 32 \mu\text{M}$.¹⁷ Building upon these knowledge, we focused on identifying conditions that allow formation of a transmembrane electron transfer chain in STEAP1. We also determined the structure of human STEAP2, which shows that it has a more flexible substrate binding site compared to STEAP1 and STEAP4.

Results

Reduction of STEAP1 by reduced FAD

We first examined if the heme on STEAP1 can be reduced by FAD. We found that reduced FAD (as FADH⁻) readily reduces STEAP1, shown by the fast decrease in the Soret absorbance of ferric heme (A_{413}) and the concomitant increase in the Soret absorbance of ferrous heme (A_{427}) (Fig. 1A). The time course of A_{427} increase is biphasic. The fast phase has a rate constant of 7.7 s^{-1} and

accounts for 60% of the total change at A_{427} , and the slow phase, 0.67 s^{-1} and 40% (**Fig. 1B**). The rate constants of both phases exhibit dependence on $[\text{FADH}^-]$. The v_{max} and K_{M} are estimated 12 s^{-1} and $4.7\text{ }\mu\text{M}$ for the fast phase, and the parameters are estimated 0.9 s^{-1} and $2.7\text{ }\mu\text{M}$ for the slow phase (**Fig. 1B**, inset).

In STEAP4 structure, a phenylalanine side chain (Phe359) is positioned between the isoalloxazine ring of FAD and the heme and is thought to mediate electron transfer from FAD to heme.¹⁵ In STEAP1-3, the equivalent residue is a leucine, Leu230 in STEAP1 (**Fig. S2A**). To examine if Leu230 is involved in electron transfer, we expressed and purified the L230G STEAP1 mutant. The UV-Vis spectrum of L230G STEAP1 is identical to that of the wild-type (WT) STEAP1 (**Fig. S2B**), indicating that the mutation does not perturb heme binding. When FADH^- was added to L230G STEAP1, biphasic reduction of heme was observed and the k 's showed dependence on $[\text{FADH}^-]$. The v_{max} and K_{M} are estimated 2 s^{-1} and $3.6\text{ }\mu\text{M}$ for the fast phase, and the parameters are estimated 0.16 s^{-1} and $1.1\text{ }\mu\text{M}$ for the slow phase (**Fig. S2C**). These results indicate that the heme in L230G STEAP1 is reduced by FADH^- more than 5 times slower than in WT STEAP1, suggesting that the side chain of Leu230 is involved in mediating electron transfer between FAD and heme in STEAPs.

Reduction of STEAP1 by cytochrome b_5 reductase

We proceeded to test whether the heme on STEAP1 can be reduced by cytochrome b_5 reductase ($b_5\text{R}$). $b_5\text{R}$ catalyzes the reduction of a tightly bound FAD by NADH and is known to reduce the heme on cytochrome b_5 (Cb_5).^{18, 19} Under anaerobic conditions, STEAP1 does not react with NADH but in the presence of both $b_5\text{R}$ and NADH, the A_{427} absorbance increases indicating reduction of the heme on STEAP1 (**Fig. S3A**). Using rapid-scan stopped-flow method, we captured the kinetics of the reactions when STEAP1 pre-incubated with $b_5\text{R}$ was mixed with NADH (**Fig. 2A**). The reduction of STEAP1 is clearly indicated by the shift of Soret peak (A_{413} to A_{427}) and the split and increase of the α and β bands at 560 nm and 530 nm, respectively (**Fig. 2A**). Three spectral species, *A*, *B*, and *C* are resolved with rate constants of $177.9 (\pm 35.3)\text{ s}^{-1}$ (*A* to *B*) and $0.13 (\pm 0.006)\text{ s}^{-1}$ (*B* to *C*), respectively (**Fig. 2A**, inset). Spectral species *A* corresponds to ferric STEAP1 plus oxidized $b_5\text{R}$ while species *C* is ferrous STEAP1 with fully reduced $b_5\text{R}$. A spectral intermediate species *B* can be identified with decreased absorbance in 420 – 500 nm range but little or no change in the Soret range when compared to *A* (**Fig. 2A**, inset), indicating that the intermediate *B* contains partially reduced $b_5\text{R}$ and a ferric heme. The resolution of intermediate *B* is due to fast FAD reduction in $b_5\text{R}$ by NADH without significant STEAP1 reduction. The *B* to *C* conversion reflects electron transfer from reduced $b_5\text{R}$ to STEAP1 (**Fig. 2A**), indicating that STEAP1 can form an electron transfer chain with $b_5\text{R}$.

We next examined whether $b_5\text{R}$ forms a complex with STEAP1. Using bio-layer interferometry assay (BLI), we measured the affinity between $b_5\text{R}$ and STEAP1 and found that $b_5\text{R}$ binds STEAP1 with a K_{D} of $\sim 5.9\text{ }\mu\text{M}$ (**Fig. S4**). We noticed that the fits to the association and dissociation profiles in BLI are rather poor at high concentrations of $b_5\text{R}$ and this is likely due to non-specific interactions between STEAP1 with $b_5\text{R}$. Nonetheless, our BLI data demonstrates that $b_5\text{R}$ can dock onto STEAP1 to establish an electron transfer chain.

We also examined reduction of L230G STEAP1 by NADH and $b_5\text{R}$ (**Fig. 2B**). Three spectral species, *A*, *B*, and *C* are resolved, and the rate constants are $78.8 (\pm 22.6)\text{ s}^{-1}$ (*A* to *B*) and $0.02 (\pm 0.01)\text{ s}^{-1}$ (*B* to *C*) (**Fig. 2B**, inset). Species *A* corresponds to ferric L230G STEAP1 with oxidized $b_5\text{R}$ while species *C* represents ferrous L230G STEAP1 with fully reduced $b_5\text{R}$ (**Fig. 2B**, inset). As in WT STEAP1, the fast reduction of $b_5\text{R}$ by NADH leads to the resolution of spectral intermediate *B*, which has decreased absorbance between 420 – 500 nm but approximately the same Soret absorbance compared to *A* (**Fig. 2B**, inset). On the other hand, the rate constant from *B* to *C*, which is electron transfer rate from reduced $b_5\text{R}$ to the heme, is significantly slower in L230G STEAP1 than in WT STEAP1 (**Fig. 2B**, inset), suggesting that Leu230 is involved in the electron transfer from $b_5\text{R}$ to STEAP1.

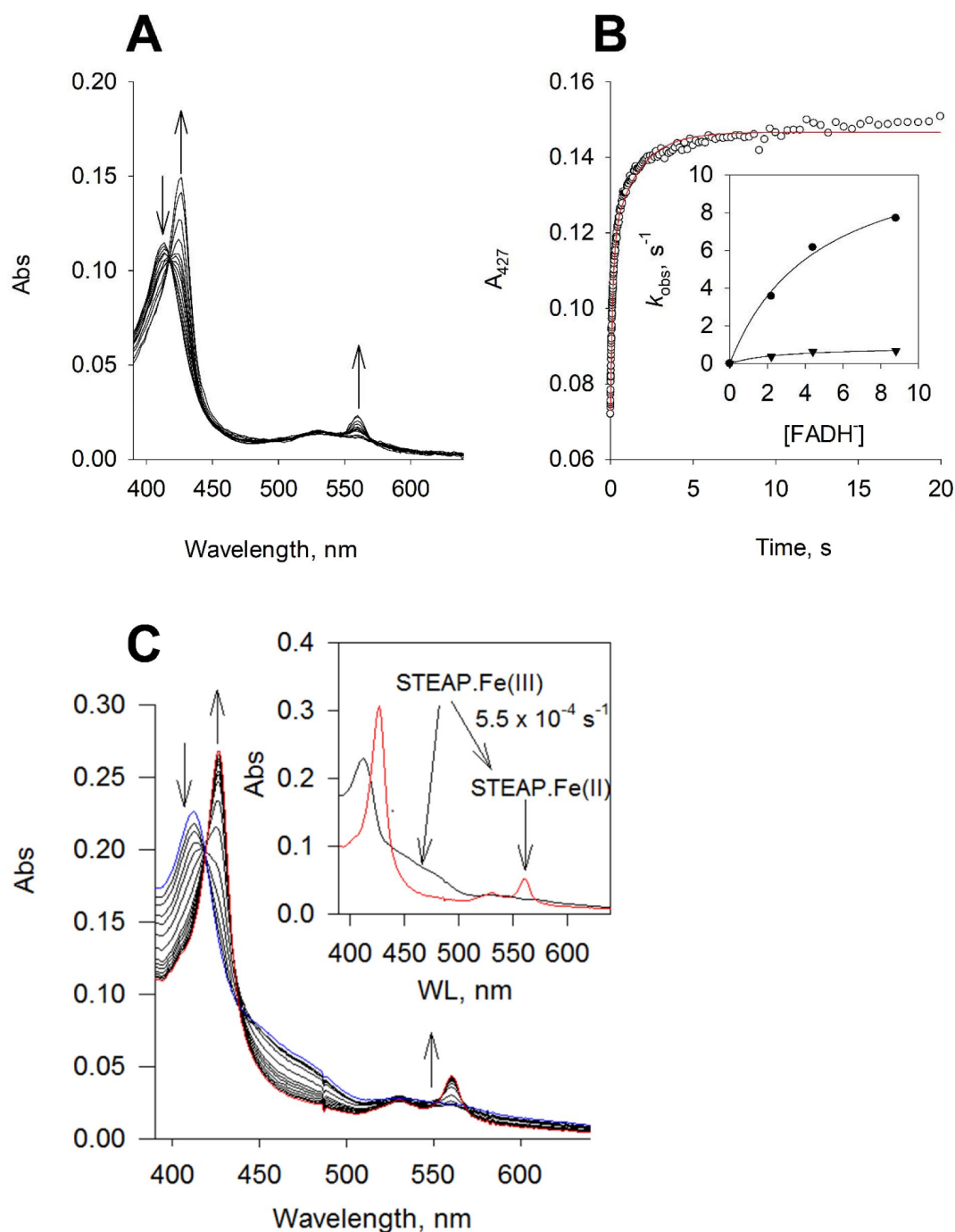


Figure 1.

Reduction of the heme on STEAP1.

(A) Rapid-scan reaction of 1.1 μM STEAP1 with 4.5 μM reduced FAD (FADH^-); the spectral change was monitored for 20 s. (B) The time course of A_{427} (black), the Soret absorbance of ferrous heme, was extracted from the rapid-scan data. Red: biphasic exponential fit with rate constants k_{obs} of $7.7 (\pm 0.30)$ and $0.67 (\pm 0.034) \text{ s}^{-1}$, respectively ($n = 3$). The percentage of each phase is 60% and 40%, respectively. Inset, the dependence of rate constants on $[\text{FADH}^-]$. Dot, the fast phase; triangle, the slow phase. Lines, fit with equation $k_{\text{obs}} = v_{\text{max}} * [\text{FADH}^-] / (K_M + [\text{FADH}^-])$. (C) The spectral changes in the reaction of a mixture of 1.1 μM STEAP2 and 0.9 μM STEAP1 (plus 2.2 μM FAD) with 60 μM NADPH; the spectral change was monitored for 1 hr. The direction of the spectral changes is indicated by the arrows. Blue, the spectrum captured at the start of the reaction; red, the spectrum after 1 hour reaction. Inset, the resolved spectral species by deconvolution and the conversion rate constant. Black, ferric STEAP and red, ferrous STEAP.

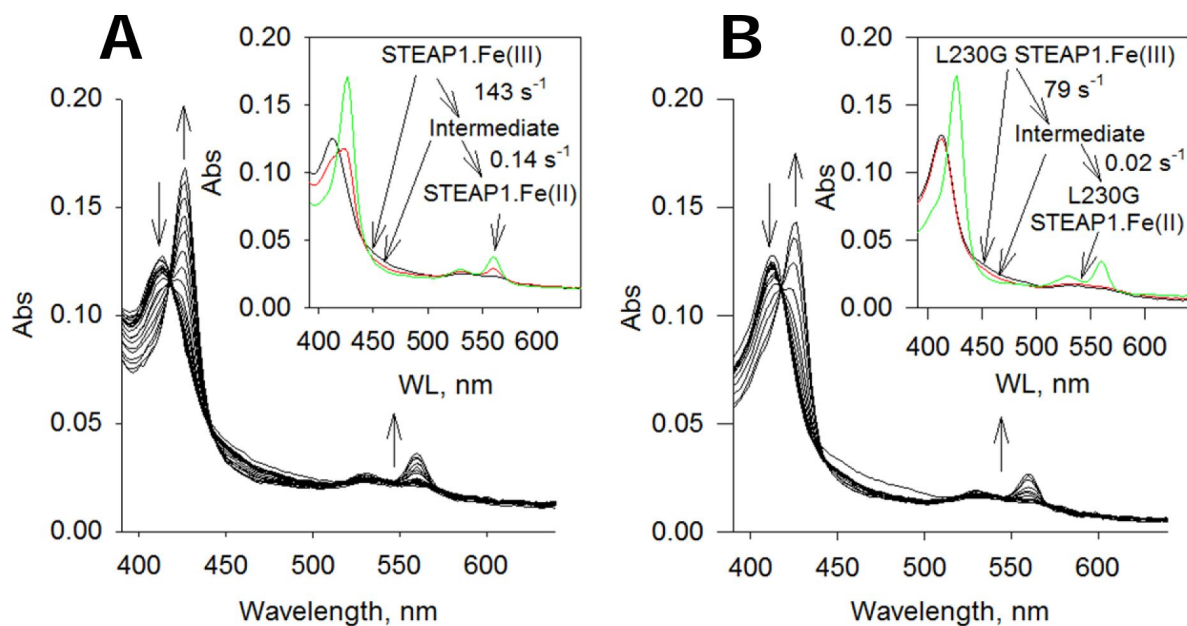


Figure 2.

Reduction of STEAP1 by cytochrome b_5 reductase.

(A) The rapid-scan reaction of 1.5 μM STEAP1 and 1.5 μM cytochrome b_5 reductase ($b_5\text{R}$) with 10 μM NADH; the spectral change was monitored for 20 s. The arrows indicate the direction of the spectral change. Inset: the resolved spectral species and the conversion rate constants. Black, ferric STEAP1 with $b_5\text{R}$, red, a spectral intermediate, and green, ferrous STEAP1 with fully reduced $b_5\text{R}$. (B) L230G STEAP1 and $b_5\text{R}$ were reacted with 10 μM NADH; the spectral change was monitored for 50 s. The direction of spectral change is indicated by the arrows. Inset: the resolved spectral species by deconvolution and the rate constants. Inset: black, ferric L230G STEAP1 with $b_5\text{R}$, red, a spectral intermediate, and green, ferrous L230G STEAP1 with fully reduced $b_5\text{R}$.

Purification and characterization of STEAP2

While structures full-length STEAP1 and STEAP4 have been determined by cryo-electron microscopy,¹⁵¹⁶ those of full-length STEAP2 and STEAP3 remain unresolved. We expressed and purified human STEAP2, which elutes as a single peak in size-exclusion chromatography and the elution volume is consistent with STEAP2 being a homotrimer (Fig. S5A). A prominent heme absorption peak is present in the purified STEAP2, and the heme content typically ranges from 70 – 90%. FAD is also detected in the purified STEAP2; however, at a level typically lower than 20% based on the fluorescence of FAD released from denatured STEAP2. No NADP(H) is detected in the purified STEAP2, suggesting that its association with STEAP2 is more transient than either heme or FAD.

The UV-Vis spectrum of STEAP2 heme are identical to that of STEAP1 heme. Ferric STEAP2 shows a Soret band at 413 nm and a broad Q band centered around 550 nm while the Soret band in ferrous STEAP2 shifts to 427 nm and the α and β bands are resolved at 560 and 532 nm, respectively (Fig. S5B). We further characterized the heme using magnetic circular dichroism (MCD) spectroscopy. The MCD spectrum of ferric STEAP2 shows strong Soret signals between 404 nm to 419 nm and no high-spin charge-transfer signal at wavelength above 600 nm while ferrous STEAP2 shows a much weaker Soret band but a very strong α band from 554 nm to 562 nm (Fig. S5C), consistent with the intense A-term Faraday effect of a typical low-spin *b*-type heme. Combined, the spectroscopic data indicates a *bis*-imidazole ligated low-spin heme in both redox states of STEAP2, consistent with a role in mediating electron transfer.

We monitored the spectral changes in STEAP2 in the reaction with NADPH. STEAP2 was pre-incubated with equal molar amount of FAD and reacted anaerobically with NADPH (Fig. S6A). In this reaction, the Soret absorbance of heme shifts from that of ferric state (A_{413}) to that of ferrous heme (A_{427}) while the α and β absorptions split into well resolved peaks at 560 nm and 532 nm, respectively, indicating the reduction of heme (Fig. S6A). Two spectral species are resolved with a transition rate constant of $1.2 (\pm 0.2) \times 10^{-3} \text{ s}^{-1}$ (*A* to *B*). Spectral species *A* corresponds to ferric STEAP2 plus FAD and spectral species *B* represents ferrous STEAP2 with reduced FAD, respectively (Fig. S6A, inset). Under the current experimental conditions, no intermediate with reduced FAD and ferric heme is resolved, suggesting that the oxidation of FAD by heme is significantly faster than its reduction by NADPH.

STEAP1 reduction by STEAP2

Following the characterization of STEAP2, we investigated whether the reduced FAD produced in STEAP2 is accessible to STEAP1. We prepared an anaerobic mixture of STEAP2 (pre-incubated with equal molar of FAD) and STEAP1 and then added NADPH (Fig. 1C). In this reaction, the Soret absorbance of heme shifts from 413 nm to 427 nm and finally a narrow peak is observed at 427 nm with no shoulder at 413 nm (Fig. 1C), indicating that the heme on both STEAP1 and STEAP2 is fully reduced. Two spectral species are resolved from the reaction of the STEAP mixture with NADPH, corresponding to ferric STEAP plus FAD and ferrous STEAP with reduced FAD, with a rate constant of $5.5 \times 10^{-4} \text{ s}^{-1}$ (Fig. 1C, inset). On the other hand, in the absence of STEAP2, only a small fraction of STEAP1 is reduced (Fig. S3B), which may come from non-enzymatic reduction of FAD by NADPH. Thus, the heme on STEAP1 can be reduced by NADPH in the presence of STEAP2, via the reduced FAD produced in the OxRD of STEAP2. This data suggests that the reduced FAD becomes diffusible and quickly finds its binding pocket in the TMD of STEAP1. Under the current experimental conditions, the rate-limiting step seems to be the production of reduced FAD in the OxRD of STEAP2. Moreover, we cannot resolve the difference between diffusion of reduced FAD from STEAP2 to STEAP1 versus repositioning of reduced FAD from the OxRD to TMD in STEAP2.

Fe³⁺-NTA reduction by STEAP1 and STEAP2

Our ability to produce reduced STEAP1 and STEAP2 allows measurement of the electron transfer step between the heme and a substrate. When reduced STEAP1 was mixed with Fe³⁺-NTA, the A₄₂₇ absorbance decreased, indicating that the ferrous heme is oxidized by Fe³⁺-NTA. The time courses of A₄₂₇ show biphasic kinetics with a fast phase accounting for 85% of the total absorbance change and a slow phase accounting for 15% (Fig. 3A). Similar biphasic kinetics was previously observed in the reactions of ferrous STEAP1 with Fe³⁺-EDTA or Fe³⁺-citrate¹⁷. We speculate that there are two STEAP1 populations in terms of the conformation of the substrate binding site or geometry of the heme, or both. The k_{on} and k_{off} rate constants of Fe³⁺-NTA are estimated based on the k_{obs} vs. [Fe³⁺-NTA] (Fig. 3B) and the K_D of Fe³⁺-NTA for the two populations of STEAP1 is calculated, 50 and 26.3 μM for the fast and slow phase, respectively (Table S1). The K_D values of Fe³⁺-NTA are similar to those of Fe³⁺-EDTA or Fe³⁺-citrate (Table S1).

The oxidation of ferrous STEAP2 by Fe³⁺-NTA is significantly slower than that of STEAP1 and the time courses of A₄₂₇ show more than two phases (Fig. 3C), suggesting more heterogeneity in the substrate binding site or heme geometry. The ΔA_{727} in the initial 2 seconds accounts for about half of the total A₄₂₇ decrease and the time courses in the initial 2 seconds can be fitted with a biphasic exponential decay function (Fig. 3C). The rate constants k_{obs} of the two phases show weak dependence on [Fe³⁺-NTA] (Fig. 3D). The k_{on} and k_{off} rate constants of Fe³⁺-NTA binding are estimated based on the k_{obs} vs. [Fe³⁺-NTA] plot (Fig. 3D) and the K_D 's of Fe³⁺-NTA are calculated, 200 and 85.7 μM for the fast and slow phases, respectively (Table S1). After the initial 2 s, the time courses of A₄₂₇ become slower with varying complicated shapes and have no clear dependence on [Fe³⁺-NTA] (Fig. 3D).

Cryo-EM structure of STEAP2

We determined the structure of STEAP2 in the presence of NADP⁺ and FAD using cryo-EM. The cryo-EM data collection, refinement, and validation statistics are summarized in Table S2. The quality of density map is sufficient to build all the major structural elements of STEAP2 *de novo* with an overall resolution of 3.2 Å (Figs. S7 and S8). The N-terminal residues 1 – 27, C-terminal residues 470 – 490, and residues 332 – 353 (loop between TM3 and 4) are not resolved in our EM map, likely due to high degree of flexibility (Fig. 4D). Similar to the structure of STEAP4, STEAP2 has a domain-swapped homotrimer structure, where the OxRD of one protomer interacts with the TMD of a neighboring protomer (Fig. 4A and 4B). Overall, the structure of STEAP2 is very similar to that of STEAP4 with a root mean squared distance (RMSD) of 0.8 Å (Cα)¹⁵.

Heme, FAD, and NADP⁺ are unambiguously resolved in the density map (Figs. 4C and S7). The FAD molecule adopts an extended conformation as observed in STEAP4¹⁵. The isoalloxazine ring buries deep in the TMD while the adenine ring of FAD forms a stacking interaction with Trp152 from the OxRD. The ribityl and pyrophosphate in FAD also interacts with residues on TMD. The isoalloxazine ring is ~10 Å away (edge-to-edge) from the heme and the side chain of Leu371 protrudes approximately midway in between (Fig. 4E). Like L230 in STEAP1 (Fig. S2A), Leu371 may mediate electron transfer in STEAP2. The distance of the isoalloxazine ring of FAD to the nearest nicotinamide ring of NADP⁺ is ~19 Å (Fig. 4E), which is too long for direct hydride transfer. We also found densities that likely correspond to phospholipid and cholesterol molecules. A 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine molecule was built between the TMDs of two neighboring protomers, and two cholesterol molecules were built on the periphery of each TMD (Fig. S7). These tightly bound lipid molecules may have relevant structural and functional roles in STEAP2.

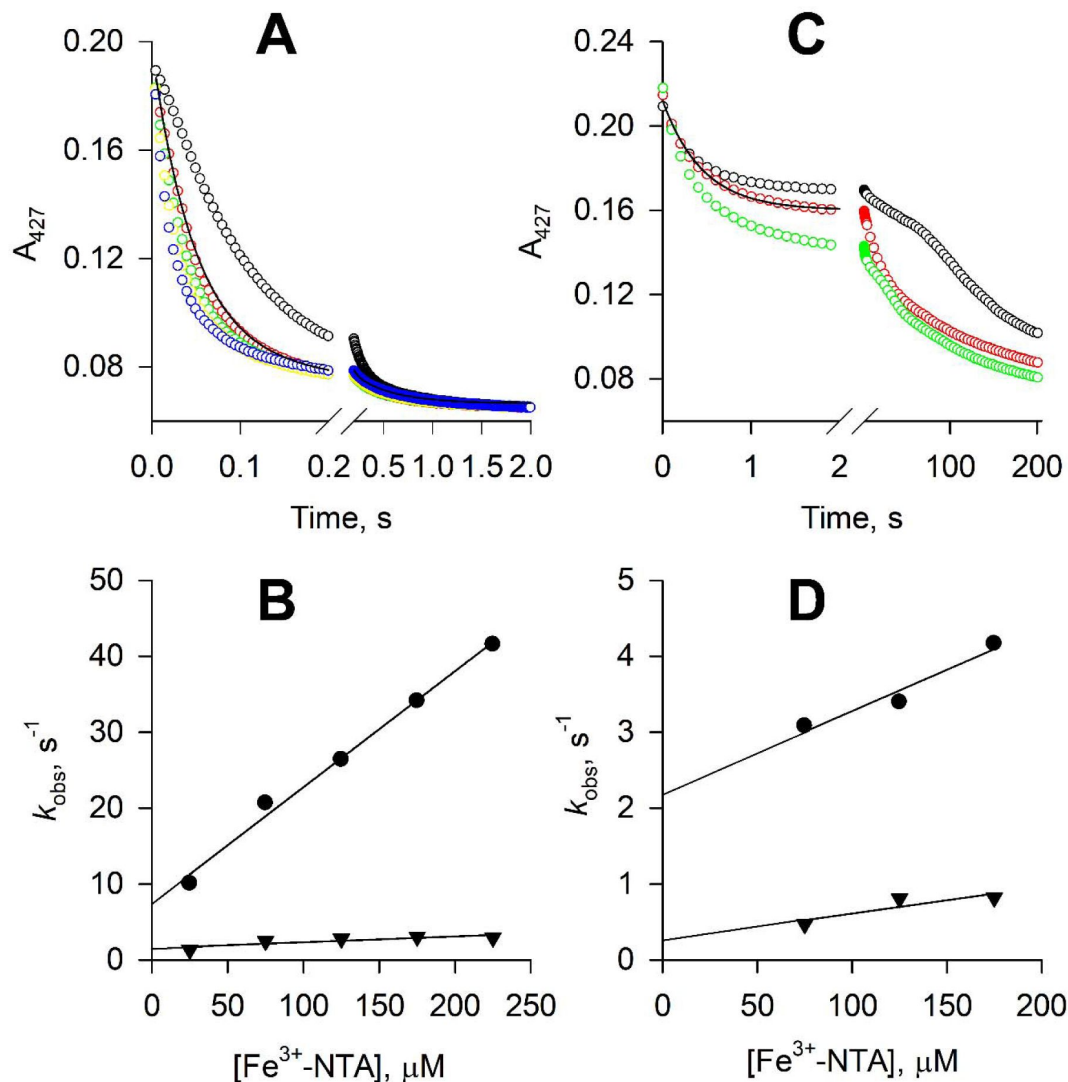


Figure 3.

Reduction of Fe^{3+} -NTA by ferrous STEAP1 and STEAP2.

(A) The time courses of A_{427} in the reactions of 1.1 μM ferrous STEAP1 with 25 (black), 75 (red), 125 (green), 175 (yellow), and 175 μM Fe^{3+} -NTA (blue). The rate constants, k_{obs} , are estimated by biphasic exponential fit to the time courses. One of such fits is shown by the black line. (B) Dependence of the rate constants k_{obs} on $[\text{Fe}^{3+}\text{-NTA}]$. Circles, k_{obs} of the fast phase of the A_{427} time courses; triangles, k_{obs} of the slow phase. (C) The time courses of A_{427} in the reactions of 1.1 μM ferrous STEAP2 with 75 (black), 125 (red), and 175 μM Fe^{3+} -NTA (green). The time courses in the initial 2 s of the reactions are fitted with a biphasic exponential function, and one such fit is shown by the black line. (D) The rate constants estimated for the initial 2 s, k_{obs} , are plotted versus $[\text{Fe}^{3+}\text{-NTA}]$. Circles, k_{obs} of the fast phase of the A_{427} time courses; triangles, k_{obs} of the slow phase. At reaction time longer than 2 s, the time courses in (C) show more complicated kinetics and no clear dependence on $[\text{Fe}^{3+}\text{-NTA}]$.

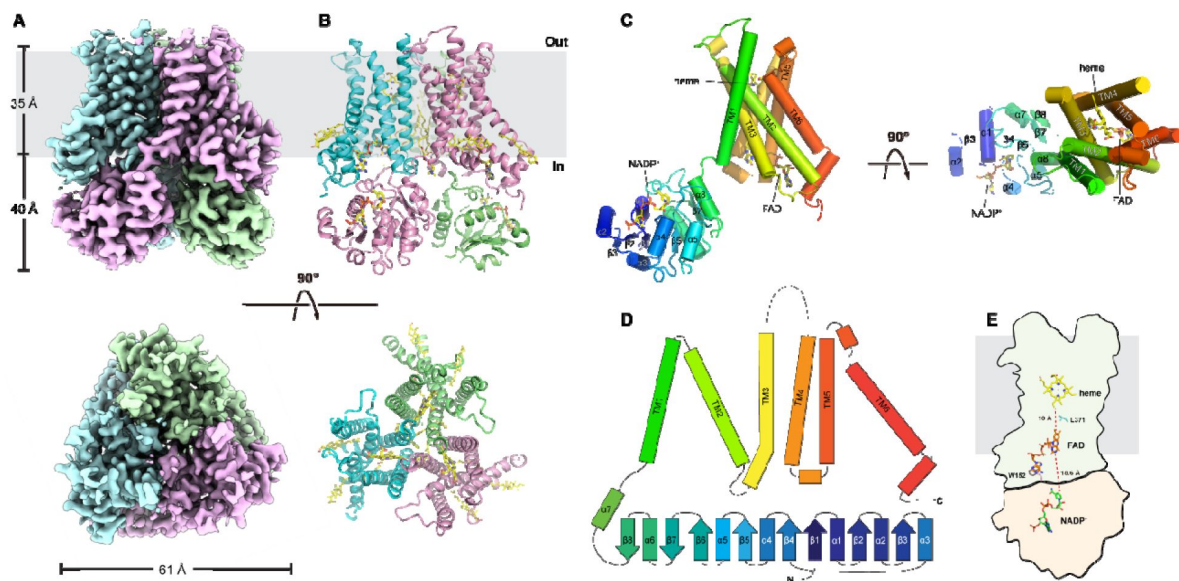


Figure 4.

Cryo-EM structure of STEAP2.

The sharpened density map (A) and cartoon presentation (B) for STEAP2 homotrimer. Top, the side view of STEAP2 homotrimer, and the grey bar represents the membrane; “in”, the intracellular side and “out”, the extracellular side. Bottom, the top view of STEAP2 homotrimer from the extracellular side. (C) The structure of one STEAP2 protomer (cartoon) with the prosthetic group heme and the cofactors FAD and NADP⁺ (sticks). Left, side view and right, top view from the extracellular side. (D) The topographic representation of the secondary structural elements. The α helices and β strands are represented by bars and arrows respectively. Dashed lines represent the unresolved segments. (E) The schematic representation of the spatial relationship of NADP⁺, FAD, and heme, shown as sticks. Trp152 and Leu371 are also shown as sticks. TMD is represented as the outline with grey shade and the OxRD with pink shade.

Reduction of STEAP2 by reduced FAD

We measured reduction STEAP2 by reduced FAD in the absence of NADPH. Reacting with 4.5 μM reduced FAD (FADH^-), ferric STEAP2 is reduced in biphasic kinetics with rates of $2.9 (\pm 0.80) \text{ s}^{-1}$ (16%) and $6.9 (\pm 0.33) \times 10^{-2} \text{ s}^{-1}$ (84%), respectively (Fig. S6B). The reduction of STEAP2 by reduced FAD is significantly slower than that of STEAP1, and we attribute this to the presence of OxRD in STEAP2, which binds to the adenosine moiety of FAD but obstruct entrance of the isoalloxazine ring of reduced FAD into the TMD.

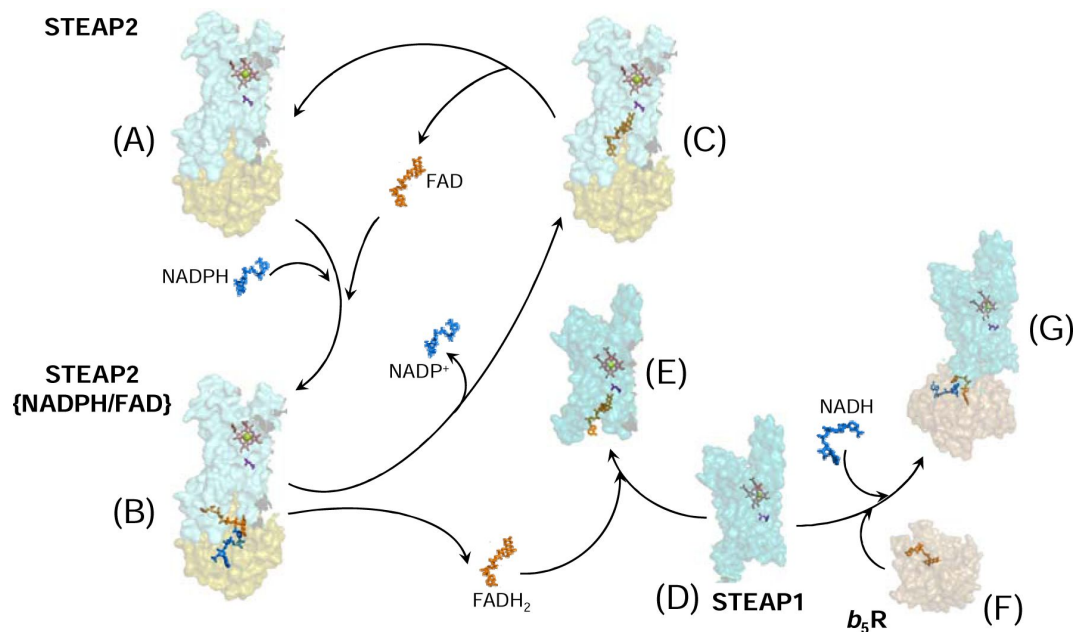
Discussion

In this study, we demonstrate that STEAP1 is reduced by reduced FAD, either supplied directly or produced in the OxRD of STEAP2. We also show that $b_5\text{R}$ can reduce STEAP1 in the presence of NADH. Thus, STEAP1 may partner with various flavin-dependent reductases to establish an electron transfer chain from either NADH or NADPH to the extracellular side. These discoveries will facilitate our understanding of the physiological functions of STEAP1.

Purified STEAP2 has low levels of bound FAD, indicating that the FAD cofactor is not tightly bound as observed in other flavin reductases such as $b_5\text{R}$ or P450 reductase. In the structure of STEAP2, we captured the bound FAD in a conformation suitable for transferring electrons to heme, and the structure is similar to that of STEAP4.¹⁵ However, the structure of STEAP2 with the bound FAD in position to receive electrons from NADPH remains unresolved. Reduction of STEAP1 in the presence of STEAP2 provides the first evidence that the bound FAD may dissociate from STEAP2 after receiving electrons from NADPH. We propose that FAD first binds to OxRD of STEAP2 in a folded conformation with its isoalloxazine ring aligned with NADPH for hydride transfer. After receiving electrons from NADPH, reduced FAD may either dissociate from the OxRD and can be utilized by STEAP2 or STEAP1 nearby or stays partially bound to the OxRD and changes into an extended conformation with the isoalloxazine ring inserted deeply into the TMD (**Scheme 1**). A “diffusible” FAD or an FAD that switches between the folded and extended conformations will inevitably limit the rate of the overall electron transfer from NADPH to the extracellular side and is consistent with the kinetics data presented in **Figs. 1C** and S6A. This mechanism is different from the electron transfer chains found in the closely related NOX or DUOX enzymes, in which the FAD likely stays bound to the OxRD and thus suitable for rapid transfer of electrons from NADPH to the extracellular side. Moreover, the mechanism of regulating the activities of STEAPs *in vivo* must be very different from those in NOX or DUOX²⁰ and warrants further investigation.

We demonstrate that STEAP1 can establish electron transfer chain with $b_5\text{R}$, which does not release FAD. We also show that $b_5\text{R}$ reduces STEAP1 from the intracellular side and it can form a complex with STEAP1. These results indicate that $b_5\text{R}$ can serve as a surrogate OxRD to complete the electron transfer chain. However, further analysis is required to establish whether STEAP1 pairs with $b_5\text{R}$ or other FAD reductase *in vivo*.

We are able to measure the rate of electron transfer from heme to a ferric substrate, and we found that STEAP2 reduces Fe^{3+} -NTA significantly slower than STEAP1. The more complex time course of heme oxidation also suggests a high level of heterogeneity in either the substrate binding pocket or the heme geometry in STEAP2. In the cryo-EM structure of STEAP2, residues 332 – 353 between TM3 and 4 are unresolved, likely due to high flexibility in this region, while in the structures of STEAP1 and STEAP4, the corresponding residues form a well-defined extracellular loop adjacent to the putative substrate binding site.^{15, 16} These differences suggest that STEAP2 has different substrates from STEAP4.



Scheme 1.

Electron transfer in STEAP1 and STEAP2.

NADPH (blue) and FAD (orange) bind to the OxRD in STEAP2 (A, olive shade) with the nicotinamide ring of NADPH aligned with the isoalloxazine ring of FAD for hydride transfer (B). The reduced FAD adopts the extended conformation with its isoalloxazine ring bound deep in the TMD of STEAP2 (C, teal shade) or dissociates from the OxRD to bind STEAP1 (D → E, cyan shade) and transfers electrons to heme (salmon). NADP(H) and FAD(H₂) are co-factors that associate with and dissociate from the STEAP protein in each redox cycle while the heme, as a prosthetic group, stays bound to the protein. Cytochrome b₅ reductase (F, sand shade) docks on STEAP1 from the intracellular side, forming a complex for electron transfer (G). The FAD-to-heme electron transfer in STEAP is likely mediated through a bulky sidechain (purple), Leu230 in STEAP1 and Leu371 in STEAP2, respectively.

Materials

4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), 5-aminolevulinic acid (5-ALA), isopropyl β -D-1-thiogalactopyranoside (IPTG), phenylmethanesulfonyl fluoride (PMSF), $\text{Fe}(\text{NO}_3)_3$, hemin chloride, dithionite, 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine (POPC), nitrilotriacetic acid (NTA), 1-ethyl-3-[3-dimethylaminopropyl]carbodiimide hydrochloride (EDC), and N-hydroxysulfosuccinimide (Sulfo-NHS) were from Sigma-Aldrich (St. Louis, MO). Lauryl maltose neopentyl glycol (LMNG) and glyco-diosgenin (GDN) were from Anatrace (Maumee, OH).

Protein expression and purification

The human STEAP2 gene (NCBI accession number AAN04080.1) was codon optimized and cloned into a modified pFastBac Dual expression vector for production of baculovirus according to the Bac-to-Bac method (Thermo Fisher Scientific, Waltham, MA). P3 viruses were used to infect High Five (*Trichoplusia ni*) or Sf9 (*Spodoptera frugiperda*) insect cells at a density of $\sim 3 \times 10^6$ cells mL^{-1} in the media including 0.5 mM 5-aminolevulinic acid, 10 μM FeCl_3 , and 5 μM hemin chloride. Infected cells were grown at 27 °C for 48 – 60 h before harvest. Cell membranes were prepared using a hypotonic/hypertonic wash protocol as previously described.²¹ Purified cell membrane pellets were then flash-frozen in liquid nitrogen for further use.

Purified membranes were thawed and homogenized in 20 mM HEPES, pH 7.5 buffer containing 150 mM NaCl, and then solubilized with 1.5% (w/v) LMNG at 4 °C for 2 h. After solubilization, cell debris was removed by ultracentrifugation (55,000 $\times$ g for 45 min at 4 °C), and hSTEAP2 was purified from the supernatant using cobalt-based Talon affinity resin (Clontech, Mountain View, CA). The C-terminal His₆-tag was cleaved with tobacco etch virus (TEV) protease at room temperature for 30 min. The protein was concentrated to around 5 mg mL^{-1} using an Amicon spin concentrator with a 100 kDa cut-off (Millipore, Burlington, MA), and then loaded onto a SRT-3C SEC-300 size-exclusion column (Sepax Technologies, Newark, DE) equilibrated with 20 mM HEPES buffer containing 150 mM NaCl and 0.01% (w/v) LMNG. For the sample used in the cryo-EM structural studies, the size-exclusion column was equilibrated with 20 mM HEPES buffer containing 150 mM NaCl and 0.02% GDN.

Rabbit STEAP1 (NCBI accession number NP_001164745.1) was expressed and purified following the method published previously.¹⁷ The L230G STEAP1 mutation was introduced by the QuikChange method (Stratagene, CA) using the primers:

forward, 5'-CGTGGGACTGGCTATCGGCGCTTTGCTGGCTGTGAC-3';

reverse, 5'-GTCACAGCCAGCAAAGCGCCGATAGCCAGTCCCACG-3'.

The cDNA of mouse cytochrome *b*₅ reductase (*b*₅R, UniProt Q3TDX8, soluble form, residues 24 – 301) was subcloned into a pET vector which appends a polyhistidine tag and a tobacco etch virus (TEV) protease site to the N-terminus of the overexpressed protein. The expression of *b*₅R followed the previous protocol²² and the cell media was supplemented with 100 μM FAD.

Electronic absorption and magnetic circular dichroism (MCD) spectroscopy

UV-Vis spectra of STEAP2 were recorded using a HP8453 diode-array spectrophotometer (Hewlett-Packard, Palo Alto, CA). The extinction coefficient of the heme was determined by pyridine hemochrome assay as published previously.¹⁷ MCD spectra of STEAP2 were recorded with a Jasco J-815 CD spectropolarimeter (Tokyo, Japan) equipped with an Olis permanent magnet

(Bogart, GA). The parameters for MCD measurements are spectral bandwidth, 5 nm; time constant, 0.5 s; scan speed, 200 nm/min. Each MCD spectrum is an average of 12 repetitive scans and the signal intensity is expressed in units of $\text{M}^{-1}\text{cm}^{-1}\text{tesla}^{-1}$.

Cryo-EM structure determination of STEAP2

Quantifoil R1.2/1.3 Cu grids were glow-discharged in air for 15 s at 10 mA in a plasma cleaner (PELCO EasiGlow, Ted Pella, Inc., Redding, CA). Glow-discharged grids were prepared using Thermo Fisher Vitrobot Mark IV. Concentrated hSTEAP2 in the presence of FAD and NADP^+ (3.5 μl) was applied to each glow-discharged grid. After blotted with filter paper (Ted Pella, Inc.) for 4.0LJs, the grids were plunged into liquid ethane cooled with liquid nitrogen. A total of 7509 micrograph stacks were collected using SerialEM^{23, 24} on a Titan Krios electron microscope (Thermo Fisher) at 300 kV with a Quantum energy filter (Gatan, Pleasanton, CA), at a nominal magnification of 105,000 $\times$ and with defocus values of $-2.5\text{ }\mu\text{m}$ to $-0.8\text{ }\mu\text{m}$. A K3 Summit direct electron detector (Gatan) was paired with the microscope. Each stack was collected in the super-resolution mode with an exposing time of 0.175 s per frame for a total of 50 frames. The dose was about 50 e^- per $\text{\AA}^2$ for each stack. The stacks were motion-corrected with MotionCor2²⁵ and binned (2×2) so that the pixel size was 1.08 $\text{\AA}$. Dose weighting²⁶ was performed during motion correction, and the defocus values were estimated with Gctf.²⁷

A total of 4,210,570 particles were automatically picked (RELION 3.1)^{28–30} from the motion-corrected images and imported into cryoSPARC³¹. After 2 rounds of two-dimensional (2D) classification, a total of 91 classes containing 1,031,895 particles were selected. A subset of 12 classes containing 117,053 particles were selected for *ab initio* three-dimensional (3D) reconstruction, producing one good class with recognizable structural features and three bad classes with no distinct structural features. Both the good and bad classes were used as references in the heterogeneous refinement (cryoSPARC) and yielded a good class at 4.10 $\text{\AA}$ from 305,849 particles. Non-uniform refinement (cryoSPARC) was then performed with C3 symmetry and an adaptive solvent mask, producing a map with an overall resolution of 3.16 $\text{\AA}$. Resolutions were estimated using the gold-standard Fourier shell correlation with a 0.143 cut-off³² and high-resolution noise substitution.³³ Local resolution was estimated using ResMap.³⁴

The structural model of STEAP2 was built based on the cryo-EM structure of STEAP4 (PDB ID: 6HCY)¹⁵ and the side chains were adjusted based on the density map. Model building was conducted in Coot³⁵. Structural refinements were carried out in PHENIX in real space with secondary structure and geometry restraints.³⁶ The EMRinger Score was calculated as described previously.³⁷

STEAP reduction by NADPH

STEAP2, 2.3 μM , or a mixture of 1.1 μM STEAP2 and 0.9 μM STEAP1 was pre-incubated with 2.5 and 2.2 μM FAD in a tonometer, respectively. The solutions were made anaerobic by 5 anaerobic cycles, each with 30 s vacuum followed by argon sparging for 4.5 min. The stock solution of NADPH was made anaerobic by N_2 sparging. Anaerobic NADPH solution was injected into the anaerobic STEAP solution using an airtight syringe and the spectral changes were monitored at room temperature using the HP 8453 spectrophotometer. The spectral changes were deconvoluted using the Pro-Kineticist program coming with the stopped-flow machine (see below).

Stopped-flow experiments

To measure the electron transfer rate from ferrous STEAP to ferric nitrilotriacetic acid (Fe^{3+} -NTA) substrate, anaerobic STEAP was first titrated to the ferrous state using dithionite and then reacted with Fe^{3+} -NTA on an Applied Photophysics model SX-18MV stopped-flow machine (Leatherhead, UK), which was placed in a COY anaerobic chamber (Grass Lake, MI). The time course of A_{427} , which reflects the oxidation of ferrous STEAP, was followed. Fe^{3+} -NTA was prepared with ferric

nitrate and NTA based on a ratio of $[\text{Fe}^{3+}]:[\text{NTA}] = 1:4$. The rate constants of the redox reactions, k_{obs} , were obtained by fitting the time courses using a monophasic or a multiphasic exponential function. The 2nd-order k_{on} rate constants of Fe^{3+} -NTA to the STEAP protein were estimated from the slopes of the linear fits to the k_{obs} vs. $[\text{Fe}^{3+}\text{-NTA}]$ plots. On the other hand, the k_{off} rate constants were estimated based on the intercepts on ordinate of the linear fits.

The anaerobic protein mixture of STEAP1 and $b_5\text{R}$ was reacted with NADH and the spectral changes were monitored using a rapid-scan diode-array accessory with the stopped-flow machine. In the reaction of STEAPs with pre-reduced FAD, FAD in 20 mM HEPES, pH7.5 containing 150 mM NaCl and 0.1% LMNG was titrated anaerobically with dithionite. Most of the reduced FAD (FADH_2) was likely in its ionized form FADH^- due its $\text{pK}_a = 6.7$. Part of the reduced FAD was re-oxidized due to the very negative potential of the FAD/FADH^- pair, and the absorbance of FAD was subtracted as the background. The spectral changes were deconvoluted using the Pro-Kineticist program coming with the stopped-flow machine.

Octet Bio-layer Interferometry

Bio-layer Interferometry (BLI) assays were performed at 30 °C under constant shaking at 1000 rpm using an Octet system (FortéBio, Fremont, CA). STEAP1 was immobilized on amine reactive second-generation (AR2G) biosensors (Sartorius, Göttingen, Germany). The biosensor tips were activated for 5 min in EDC and 10 mM Sulfo-NHS before being loaded with STEAP1 at a concentration of 1 $\mu\text{g}/\text{mL}$ for 10 min. The tips were then quenched in 1 M ethanolamine (pH 8.5) for 5 min and equilibrated in 20 mM HEPES, pH7.5 containing 150 mM NaCl, 0.1% LMNG, and 0.1% BSA to reduce non-specific binding. The tips were then transferred into wells containing various concentration of $b_5\text{R}$, 20, 10, 5, 2.5, 1.3, and 0.6 μM , for association and then back to the equilibration wells for dissociation. The binding curves were aligned and corrected with the channel with no analyst protein. The association and dissociation phases were fitted with a monophasic exponential function. The equilibrium responses (Req) in the association incubation were plotted against $[b_5\text{R}]$ and fitted with a dose-response function to calculate the dissociation constant K_D of STEAP/ $b_5\text{R}$ complex.

Data and Materials Availability

The EM data and fitted model of human STEAP2 are deposited in the Electron Microscopy Data Bank (EMD-25775) and the RCSB Protein Data Bank (7TAI). Original data is available upon request, please contact gang.wu@uth.tmc.edu.

Author Contribution Statement

GW and MZ initiated and designed the project. KC and LW led the effort of protein production and structure determination and with help from JS and GW. JS prepared protein mutant and measured protein-protein binding affinity. GW designed and conducted spectroscopic measurements. All authors analyzed and interpreted the data. GW and MZ wrote the manuscript with input from all authors.

Competing Interests

The authors declare no competing interests.

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Reviewer #1 (Public Review):

In the revised manuscript presented by Chen, Wang, and coworkers, the authors examine two proteins, STEAP1 and STEAP2, which are transmembrane hemoproteins that are involved in Fe and Cu homeostasis and are implicated in certain cancer states. The authors produce recombinant forms of STEAP1 and STEAP2 and attempt to reconstruct the electron-transport chains of both; under certain conditions, the electron transport chain of STEAP2 consists of an internal reductase domain that binds NADPH and transfers electrons to an internal FAD molecule prior to the heme b, while STEAP1 can use an independent/external b5 reductase instead of an intrinsic reductase domain to accomplish the same electron transport pathway. A strong feature of this manuscript is the determination of the cryo-EM structure of the human STEAP2 protein resolved to 3.2 Å globally and bound to heme, FAD (in an extended conformation), and NADP⁺/NADPH.

This revised study aims to address the previous weaknesses that were noted, such as the unclear presentation of the kinetics data, the lack of determined redox couples, the lack of in vivo oligomerization verification, and some minor weaknesses such as the fit of the BLI data and the exact redox states of the bound coenzymes. In general, the authors have sought to rectify these weaknesses chiefly through textual edits. Through these revisions, the kinetics data are now better presented and may be more easily interpreted by the reader, how the samples for cryo-EM were prepared with the respective coenzymes is clearer, and a comparison between the oligomerization of STEAP2 and STEAP4 suggests conservation of oligomerization. The determination of the redox potentials of the hemes in both STEAP1 and STEAP2 would still be a strong addition to the data presented, but it is recognized that the limitations in the ability to prepare sufficient quantities of recombinant enzyme limits the ability to determine the measurements and may represent another publication outside of the scope of this publication.

Reviewer #2 (Public Review):

Human STEAPs form a family of transmembrane heme-bound proteins. They are implicated in cancer given their high expression levels in tumor cells. Previous work has revealed that STEAPs 1-4 are iron and copper reductases. The recent structure determination of STEAP1 and STEAP4 unveiled their trimeric arrangement. STEAP1 is an outlier because it lacks the cytosolic reductase domain present in STEAPs 2-5. The present work adds to our knowledge of the family. It reports on the cryoEM structure of STEAP2 that is similar to the known structures of STEAP4 and STEAP1. The structural analysis provides additional support to a FAD-dependent heme-reduction mechanism whereby FAD oscillates between two conformations. The excellent kinetics experiments show that STEAP1 can be promiscuous regarding the source of electron donors that it can use. Indeed, cytochrome b5 can directly reduce the heme prosthetic group of STEAP1 thereby establishing an electron transfer chain that conveys electrons from NAD(P)H to the extracellular iron. Remarkably, STEAP1 can also accept electrons from free reduced FAD. Most interestingly, the manuscript demonstrates that STEAP2 can be a source of reduced FAD so that STEAP2 can create the reducing power needed for its own activity and the activity of STEAP1. This work further convincingly shows that STEAP1 can reduce iron whereas STEAP2 is less effective in iron reduction. The manuscript indicates that STEAP2 might accept other substrates providing a hint about the distinct biochemical and physiological roles of the STEAP paralogs. The manuscript does not address this point that remains open for further investigations. Aside from this minor weakness, the manuscript will advance the fields of STEAP and iron biochemistry. It has benefited from the advice given by the Reviewers leading to a high-quality presentation and data analysis.

Reviewer #3 (Public Review):

The six-transmembrane epithelial antigen of the prostate (STEAP) family comprises four members in metazoans. STEAP1 was identified as integral membrane protein highly upregulated on the plasma membrane of prostate cancer cells (PMID: 10588738), and it later became evident that other STEAP proteins are also over expressed in cancers, making STEAPs potential therapeutic targets (PMID: 22804687). Functionally, STEAP2-4 are ferric and cupric reductases that are important for maintaining cellular metal uptake (PMIDs: 16227996, 16609065). The cellular function of STEAP1 remains unknown, as it cannot function as an independent metalloredoxase. In the last years, structural and functional data have revealed that STEAPs form trimeric assemblies and that they transport electrons from intracellular NADPH, through membrane bound FAD and heme cofactors, to extracellular metal ions (PMIDs: 23733181, 26205815, 30337524). In addition, numerous studies (including a previous study from the senior authors) have provided strong implications for a potential metalloredoxase function of STEAP1 (PMIDs: 27792302, 32409586).

This new study by Chen et al. aims to further characterize the previously established electron transport chain in STEAPs in high molecular detail through a variety of assays. This is a well-performed study that provides new insights into the established mechanism of electron transport in STEAP proteins. The authors first perform a detailed spectroscopic analysis of STEAP1, and present the interesting observation that STEAP1 can receive electrons from cytochrome b5 reductase. Then, a similar spectroscopic analysis is performed on another STEAP family member, STEAP2, followed by experiments that show how reduced FAD can diffuse from STEAP2 to STEAP1 to reduce the heme of STEAP1. Finally, the cryo-EM structure of STEAP2 is presented.

Experimentally, the conclusions are appropriate and consistent with the experimental data. The observation that STEAP1 can form an electron transfer chain with cytochrome b5 reductase in vitro is an exciting finding, but its physiological relevance remains to be validated. The metalloredoxase activity of STEAP1 in vitro has been described previously by

the authors and by others (PMIDs: 27792302, 32409586). However, when over expressed in HEK cells, STEAP1 by itself does not display metal ion reductase activity (PMID: 16609065), and it was also found that STEAP1 over expression does not impact iron uptake and reduction in Ewing's sarcoma (cancer) cells (PMID: 22080479). Therefore, the physiological relevance of metal ion reduction by STEAP1 remains controversial. Future studies will have to elucidate if the established interaction between STEAP1 and cytochrome b5 reductase is relevant in cells.

The work will be interesting for scientists working within the STEAP field and for those working on other oxidoreductases. The spectroscopic data is robust and However, the new structural insights into STEAP2 are limited because the structure is virtually identical to the published structures of STEAP4 and STEAP1 (PMIDs: 30337524, 32409586), which is not surprising because of the high sequence similarity between the STEAP isoforms. When taken together, this study by Chen et al. strengthens and supports previously published biochemical and structural data on STEAP proteins, making an important contribution to the STEAP field.

Author Response

The following is the authors' response to the original reviews.

We appreciate the critical review of our manuscript. We believe that we have addressed the questions and concerns raised by the reviewers to the best of our ability. As part of the revision, we conducted two new experiments to enhance the rigor of the conclusions and to provide more insights into the mechanism of STEAP proteins, and we reorganized the Results section, as suggested by the reviewers, following to a clearer logical thread. The new data are briefly summarized below.

1. Reduction of L230G STEAP1 by reduced FAD. We made Leu230Gly STEAP1 mutant and measured the rate of heme reduction by reduced FAD. We found that the rate of heme reduction in L230G STEAP1 is slower than that in the wild type STEAP1. Since Leu230 is solvent accessible only from the intracellular side, this result supports the conclusion that reduced FAD binds to STEAP1 on the intracellular side and reduces the heme. This result also indicates that leucine, which is found at the equivalent position in STEAP1, 2 and 3, and Phe359 in STEAP4, has a significant role in mediating electron transfer from FAD to the bound heme.
2. Reduction of STEAP2 by reduced FAD. We showed that STEAP2 can be reduced when supplied with reduced FAD, and that the rate of heme reduction is significantly slower than that of reduction of STEAP1 by reduced FAD. This result is consistent with presence of the oxidoreductase domain (OxRD)[†] in STEAP2, which hampers direct entrance of the isoalloxazine ring of FAD to its binding pocket in the transmembrane domain (TMD). On the other hand, the rate of heme reduction by reduced FAD is much faster than that of heme reduction in the presence of NADPH and FAD, indicating that reduction of FAD by NADPH is rate-limiting in the electron transfer chain in STEAP2.

[†]: To be consistent with literature, we adopted the nomenclature “oxidoreductase domain (OxRD)” for the N-terminal soluble domain in STEAP proteins. We used the term “reductase domain (RED)” in the previous version of our manuscript.

Reviewer #1 (Public Review):

This important study reveals the structure of human STEAP2 for the first time and suggests the electron transport pathway, but some questions remain regarding the interpretation of the in vitro electron transport experiments, the lack of available redox

couples, and potential alternative hypotheses that would if addressed, strengthen the claims in the manuscript.

Strengths

One of the clear strengths of the manuscript that stands out is the determination of the structure of human STEAP2. The structures of some other homologs are known, but STEAP2's structure was not, and STEAP2 seems to have an unusually low activity towards certain metal chelates. The approach of producing the human STEAP2 in insect cells with the supplementation of cofactor biogenesis components nicely results in cofactor-replete protein. The structure of STEAP2 reveals a domain-swapped trimer, with the NADPH-binding domain of the neighboring protomer within electron-transport distance of the FAD-heme axis. The FAD has an interesting and somewhat unusual extended conformation and abuts a Leu residue that may regulate electron transport. Another strength of the manuscript is the demonstration that STEAP1, which does not have the internal NADPH binding domain, can interact modestly and shuttle electrons to the heme in STEAP1 through FAD. These experiments nicely expand information on the function of STEAP1 and provide a structural basis for electron transport in STEAP2.

Weaknesses

A major weakness in the manuscript lies with the kinetics data and how the data, as presented, are unclear to the reader regarding their impact and their connection to the purported electron transport scheme. While multiple sets of data are reported, the analysis in all cases is simply the reduction of a hexacoordinate heme and its related spectra and kinetic parameters. In most cases, it's unclear to the reader which part of the electron pathway is being tested in which experiment. Simple diagrams would be helpful in each case. However, it's also unclear if all of the potential order of addition experiments were actually performed; i.e., flavin but no NADPH; NADPH but no flavin; flavin before NADPH; flavin after NADPH, etc. As there are multiple permutations that should be tested to demonstrate the electron transport pathway, presenting the data in a way that makes it clear to the reader is challenging. Particularly missing are the determined redox potentials of the hemes in both STEAP1 and STEAP2. Could differences in these heme redox potentials be drivers of the difference in metal reduction rates?

We re-structured the manuscript to follow a clearer logical thread. We provided explanations for which electron transfer steps are being examined in each experiment.

We cannot reliably determine EM due to insufficient amount of purified proteins. We are inclined to think that the bound heme on STEAP1 and STEAP2 have similar EM, based on their similar coordination geometry and nearly identical UV-Vis and MCD spectra. Thus, different rates of Fe³⁺-NTA reduction by STEAP1 and STEAP2 are likely due to differences in substrate binding site rather than different EM.

Also, the text indicates that STEAP2 does not show a reduction rate dependence on the [Fe³⁺+NTA], but Figure 1A shows a difference in rates dependent on [Fe³⁺+NTA], and the shape of the reduction curve also changes based on [Fe³⁺+NTA]. This discrepancy should be rectified.

We fixed this error. The reduction of Fe³⁺-NTA by ferrous STEAP2 shows multiple phases and the reaction rates within the initial 2 seconds are weakly dependent on [Fe³⁺+NTA].

A second major weakness is the lack of any verification of the relevance of the STEAP2 oligomerization to its *in vivo* function. Is the same domain-swapped trimer known to exist *in vivo*? If the protein were prepared in other detergents, is the oligomerization

preserved? It is alluded to in the text that another STEAP protein is also a trimer. Was this oligomerization verified in vivo?

The domain-swapped assembly is an interesting phenomenon, and it seems to provide a solution for bringing the FAD closer to heme. The same domain swapped trimeric assembly is also observed in the structure of STEAP4, which was purified in a different detergent (Nat Commun (2018), 9, page 4337). It is likely that this feature is shared by STEAP2, 3, and 4, and preserved during the purification process.

Could this oligomerization be disrupted to impede or abrogate electron transport to underscore the oligomerization relevance? This point is germane, as it would further suggest that the domain-swapped trimer observed in the STEAP2 cryo-EM structure is physiologically relevant, especially given how far the distance between the NADPH and the FAD would otherwise be to support electron transport.

We agree with the reviewer's reasoning that the oligomeric assembly is required for proper function of STEAPs and thus could potentially be utilized for functional regulation. However, we are not aware of any physiologically relevant stimuli or treatment that would allow regulation of STEAP functions by inducing or forming different oligomeric states. Our experience with STEAP proteins is that the trimeric assembly is stable and well-preserved during the purification process and can only be disrupted under denaturing conditions such as SDS-PAGE.

Beyond these two areas in which the manuscript could be improved there are a couple of minor considerations. First, the modest resolution of the STEAP2 structure prevents assigning the states of NADP⁺/NADPH and FAD/FADH₂ with confidence. An orthogonal measure would be useful for modeling the accurate states in the structure.

We agree. We clarified the ambiguity and stated in the main text that the cryo-EM structure of STEAP2 was determined in the presence of NADP⁺ and FAD.

Finally, the BLI b5R/STEAP1 binding/unbinding fits seem somewhat poor, especially at high concentrations of b5R in the dissociation regime, which likely influences the derived value of K_d. A different fitting equilibrium might yield better agreement between the experimental and theoretical results. Moreover, whether this binding strength is influenced by the reduction state of the NADPH would be helpful in understanding and contextualizing the weak binding affinity.

We think that non-specific binding likely causes deviations from the simple binding model at higher b5R concentrations. We made a comment on this in the main text. We agree with the reviewer that the interactions between b5R and STEAP1 could be redox dependent, for example, a reduced FAD on b5R may enhance the affinity. We could implement this by performing the binding experiments in an anaerobic chamber, but this is beyond the scope of the current study.

Reviewer #2 (Public Review):

The manuscript provides new insight into a family of human enzymes. It demonstrates that STEAP2 can reduce iron and STEAP1 can be promiscuous regarding the source of electron donors that it can use. The quality of the kinetics experiment and the structural analysis is excellent. I am less enthusiastic about the interpretation of data and the take-home message that the manuscript intends to deliver. Above all, the work combines data on STEAP2 and STEAP1 and it remains unclear which questions are ultimately addressed. A second critical point is about the interpretation of the experiment demonstrating that

STEAP1 can be reduced by cytochrome b5 reductase. The abstract states that "We show that STEAP1 can form an electron transfer chain with cytochrome b5 reductase" whereas the main text discusses the data by suggesting that "we speculate that FAD on b5R may partially dissociate to straddle between the two proteins". The two statements seem to be partly contradictory. Cytochrome b5 reductases do not easily release FAD but rather directly donate electrons to heme-protein acceptors (PMID: 36441026). According to the methods section, no FAD was added to the reaction mix used for the cytochrome b5 reductase experiment. Overall, the data seem to indicate that the reductase might reduce the heme of STEAP1 directly. Would it be possible to check whether FAD addition affects the kinetics of the process?

We agree with the reviewer on this point. We do not have evidence indicating that FAD fully or partially dissociates from b5R to interact with STEAP1. We removed the statement in the revision.

We have not tried to add free reduced FAD to the mixture of STEAP1/b5R/NADH, because we feel that this would increase the complexity of the system and complicate data interpretation. We are working on determining the structure of b5R in complex with STEAP1 to visualize the electron transfer pathway, and we hope that such a structure would provide a framework for understanding electron transfer between the two proteins.

And to perform a control experiment to check that NAD(P)H does not directly reduce the heme of STEAP1 (though unlikely)?

We did the control experiment and included data in Fig. S3A. No reduction of heme by NADH alone.

A final point concerns the "slow Fe3+-NTA reduction by STEAP2". The reaction is not that slow as the initial phase is 2 per second. The reaction does not show dependence on the substrate concentration suggesting "poor substrate binding". I am not convinced by this interpretation. Poor substrate binding would give rise to substrate dependency as saturation would not be achieved. A possible interpretation could be that substrate binding is instead tight and the enzyme is saturated by the substrate. Can it be that the reaction is limited by non-productive substrate-binding and/or by interconversions between active and non-active conformations? We re-analyzed the data and revised the Results and Discussion.

We agree with the reviewer on this point. We re-analyzed the data and found that the reaction rates within the first 2 seconds are weakly dependent on [Fe3+-NTA] while the rates beyond 2 seconds do not show dependence on [Fe3+-NTA]. More studies are required to unravel the mechanism that leads to the complicated kinetic data.

Reviewer #3 (Public Review):

The six-transmembrane epithelial antigen of the prostate (STEAP) family comprises four members in metazoans. STEAP1 was identified as integral membrane protein highly upregulated on the plasma membrane of prostate cancer cells (PMID: 10588738), and it later became evident that other STEAP proteins are also over expressed in cancers, making STEAPs potential therapeutic targets (PMID: 22804687). Functionally, STEAP2-4 are ferric and cupric reductases that are important for maintaining cellular metal uptake (PMIDs: 16227996, 16609065). The cellular function of STEAP1 remains unknown, as it cannot function as an independent metalloredutase. In the last years, structural and functional data have revealed that STEAPs form trimeric assemblies and that they transport electrons from intracellular NADPH, through membrane bound FAD and heme

cofactors, to extracellular metal ions (PMIDs: 23733181, 26205815, 30337524). In addition, numerous studies (including a previous study from the senior authors) have provided strong implications for a potential metalloredox function of STEAP1 (PMIDs: 27792302, 32409586).

This new study by Chen et al. aims to further characterize the previously established electron transport chain in STEAPs in high molecular detail through a variety of assays. This is a wellperformed, highly specialized study that provides some useful extra insights into the established mechanism of electron transport in STEAP proteins. The authors first perform a detailed spectroscopic analysis of Fe³⁺-NTA reduction by STEAP2 and STEAP1, confirming that both purified proteins are capable of reducing metal ions. A cryo-EM structure of STEAP2 is also presented. It is then established that STEAP1 can receive electrons from cytochrome b5 reductase, and the authors provide experimental evidence that the flavin in STEAP proteins becomes diffusible.

The specific aims of the study are clear, but it is not always obvious why certain experiments are performed only on STEAP2, on STEAP1, or on both isoforms. A better justification of the performed experiments through connecting paragraphs and proper referencing of the literature would improve readability of the manuscript. Experimentally, the conclusions are appropriate and mostly consistent with the experimental data, although one important finding can benefit from further clarification. Namely, the observation that STEAP1 can form an electron transfer chain with cytochrome b5 reductase in vitro is an exciting finding, but its physiological relevance remains to be validated. The metalloredox activity of STEAP1 in vitro has been described previously by the authors and by others (PMIDs: 27792302, 32409586). However, when over expressed in HEK cells, STEAP1 by itself does not display metal ion reductase activity (PMID: 16609065), and it was also found that STEAP1 over expression does not impact iron uptake and reduction in Ewing's sarcoma (cancer) cells (PMID: 22080479). Therefore, the physiological relevance of metal ion reduction by STEAP1 remains controversial. The current work establishes an electron transfer chain between STEAP1 and cytochrome b5 reductase in vitro with purified proteins. However, the conformation of this metalloredox activity of the STEAP1-cytochrome b5 complex will be required in a cell line to prove that the two proteins indeed form a physiological relevant complex and that the results are not just an in vitro artefact from using high concentrations of purified proteins.

The work will be interesting for scientists working within the STEAP field. However, some of the presented data are redundant with previous findings, moderating the study's impact. For instance, the new structural insights into STEAP2 are limited because the structure is virtually identical to the published structures of STEAP4 and STEAP1 (PMIDs: 30337524, 32409586), which is not surprising because of the high sequence similarity between the STEAP isoforms. Moreover, the authors provide experimental evidence to prove the previous hypothesis (PMID: 30337524) that the flavin in STEAP proteins becomes diffusible, but the molecular arrangement of a STEAP protein, in which the flavin interacts with NADPH, remains unknown. Based on the manuscript title, I would also expect the in-depth characterization of STEAP1/STEAP2 hetero trimers (first identified by the authors), but this is only briefly mentioned. When taken together, this study by Chen et al. strengthens and supports previously published biochemical and structural data on STEAP proteins, without revealing many prominent conceptual advances.

We thank the reviewer for information and the broader context. We have revised the manuscript to have a clearer logical thread.

Reviewer #1 (Recommendations For The Authors):

Please see the "Public Review" for recommendations.

Reviewer #2 (Recommendations For The Authors):

Specific suggestions

-The introduction should more clearly state which questions are being addressed and why STEAP1 and STEAP2 are investigated.

We have revised the Introduction to make that clearer.

-The manuscript should discuss more extensively and provide possible explanations for the substrate-independent kinetics of iron-reduction by STEAP2.

We re-analyzed the data and found the rate constants of the reactions before 2 s are weakly [Fe3+NTA]-dependent. We ascribe the weak [Fe3+-NTA]-dependence to the partial rate-limiting by substrate binding. However, we do not have a good interpretation for the reaction kinetics after 2 s which does not show [Fe3+-NTA]-dependence.

-"The rate of STEAP1(Fe(II)) oxidation by Fe3+-NTA is similar to those by Fe3+-EDTA or Fe3+-citrate, but the rates are significantly faster than STEAP2(Fe(II)) re-oxidation by Fe3+NTA (Fig. 1B)." The rates for STEAP1 should be given to substantiate this statement.

We added Table S1 in the supplementary materials that includes the 2nd order association (k_{on}) and the 1st order dissociation rate constants (k_{off}) of iron substrates in STEAP1 and STEAP2. Data on Fe3+-EDTA or Fe3+-citrate by STEAP1 are from our previous study (Biochemistry, 2016). We also calculated the KDs of different iron substrates for STEAP1 and STEAP2.

- "Our results indicate that STEAP2 can supply reduce FAD to initiate electron transfer in STEAP1." As discussed above, this statement should be discussed and analyzed*

We mixed 0.9 μ M STEAP1, 1.1 μ M STEAP2, and 2.2 μ M FAD and added 60 μ M NADPH to the system and found that the heme on both STEAP1 and STEAP2 are reduced. Since adding NADPH to STEAP1 plus FAD alone does not reduce the heme (Fig. S3B), we reasoned that reduction of the heme on STEAP1 is achieved by the reduced FAD produced on STEAP2. The reduced FAD likely dissociates from STEAP2 and then bind to STEAP1.

-Experiments on "STEAP1 reduction by STEAP2" The experiments show that "STEAP2 can supply reduce FAD to initiate electron transfer in STEAP1." Could these results be explained by heterotrimer formation in agreement with the previous data published by the authors?

In our experience, STEAP1 and STEAP2 homotrimers are stable and do not form heterotrimers when mixed. STEAP1/2 heterotrimers form only when the two proteins are co-expressed in cells (Biochemistry (2016) 55, 6673-6684).

Reviewer #3 (Recommendations For The Authors):

Major points:

1. As a very general point: the order in which the results are presented could be greatly improved to increase the readability for non-experts. To elaborate: The manuscript starts with the spectroscopic characterization of STEAP2, then suddenly the reductase activities of STEAP1 and STEAP2 are compared; subsequently, experiments are described involving STEAP1 and cytochrome b5 reductase; then the cryo-EM structure of STEAP2 is presented etc. As a non-expert reader, this presentation of the results is confusing, especially because the paragraphs are not always connected well, and there is a lot of switching between STEAP1 and STEAP2 data. A more logical order would be to first present the STEAP2 spectroscopy and structural data, then write a connecting paragraph on why it is important to also study the electron transfer chain in STEAP1, followed by the comparison of the STEAP isoforms and the data on STEAP1 alone. The authors should include sentences on why they performed each experiment. For example, why did they determine the structure of STEAP2. What were they after that they could not retrieve from the homologous STEAP4 and STEAP1 structures? Justification of the performed experiments will make it easier for the reader, and will establish a better connection between the paragraphs.

We reorganized the data presentation in Results per the reviewer's suggestions.

1. The physiological relevance of metal ion reduction by STEAP1 remains controversial. Because the current work establishes an electron transfer chain between STEAP1 and cytochrome b5 reductase, could the authors perform an easy experiment where they over express both STEAP1 and cytochrome b5 reductase in a cell line? If such an experiment would reveal STEAP1-dependent metal-ion reduction, it would greatly improve this part of the manuscript. If no activity is observed, the established electron transfer chain could just represent an in vitro artifact from using high concentrations of purified proteins.

This is an excellent point. We are not set up to perform the proposed experiment but will do so in the future.

1. The manuscript states that metal ion reduction of purified STEAP2 is slow, and the authors explain this by the absence of density for the extracellular region between helices 3 and 4 that are present in the structures of STEAP4 and STEAP1, resulting in a less-well defined substrate-binding site. Can the authors exclude that the less-well defined substrate-binding site is a result of the detergent extraction of STEAP2? Would it be possible to measure the reductase activity of STEAP2 in purified membranes?

Detergent mostly interacts with the transmembrane domains and since the TMD region of STEAP2 aligns well with those of STEAP1 and STEAP4, we do not think that the disordered substrate binding region in STEAP2 is a consequence of detergent solubilization. It is difficult to conduct pre-steady state kinetic experiments using STEAP2 in membrane fractions.

1. The manuscript would greatly benefit from citing the literature more comprehensively to acknowledge insightful findings from authors in the field; for example, the important work by the Lawrence lab from 2015 (PMID: 26205815), which biochemically proved that STEAPs bind a single heme and that FAD bridges the TMD and RED, is not cited. The authors also mention that STEAP proteins belong to the same family as NOX proteins and cite some NOX structure papers. However, they fail to cite the first NOX structure paper (PMID: 28607049), as well the manuscript that structurally compares NOXs and STEAPs (PMID: 32815713). Similarly, the authors use SerialEM for their cryo-EM data collection but cite an old paper instead of the more recent (and relevant) SerialEM publication (PMID: 31086343).

We agree and added the references.

1. Generally, the data presented in the manuscript appear of good technical quality. However, a 'Table 1' with all relevant cryo-EM data collection and refinement statistics is completely missing as far as I can see. The authors should definitely add this to allow for the judgement of structural data quality. Without it, the manuscript is not suitable for publication.

We added Table S2 that includes relevant cryo-EM statistics.

Minor points:

1. The authors write in the abstract: 'STEAP2 - 4, but not STEAP1, have an intracellular domain that binds to NADPH and FAD'. This is not correct, because it has clearly been established that FAD also majorly binds to the transmembrane domain (this is even shown by the authors in the current manuscript as well).

Agree, we corrected that in the revision.

1. Sentence from the abstract and introduction state: 'It is also unclear whether STEAP1 has metal ion reductase activity' and 'it is unclear whether STEAP1 can form a competent electron transfer chain from NADPH'. The authors should definitely add "physiologically relevant" to these sentences. Namely, the senior authors themselves concluded in their 2016 Biochemistry paper (PMID: 27792302) that STEAP1 is capable of reducing metal ion complexes. Further indications that the transmembrane domain of STEAP1 displays metalloredutase activity was published by the Gros lab (PMID: 32409586), and it was also shown that fusing the RED of STEAP4 to the TMD of STEAP1 yields a functional protein in cells that reduces metal ions.

Good point and we revised the text and included the references.

1. Why is scheme 1 not just a summarizing figure?

We could change Scheme 1 to a Figure if required by the copy editor.

1. What is the purpose of Fig. 6? A larger depiction of Fig. 5e would likely be more relevant and should be considered as a replacement. Alternatively, the structure of STEAP1 (pdb 6y9b) could be shown in combination with Fig. 7, as the mutation is performed in STEAP1.

We agree and made changes to the structural figures to enhance clarity.

1. The manuscript now contains many, single panel figures. Certain main figures could easily be combined, for example, Fig. 1 and 2 and/or Fig. 3 and 4.

We agree and made changes to reduce single panel figures.

1. In Fig. 2, 3 and 4, the spectra show changes in peak heights as a result of the ferric to ferrous heme transition. However, a time component is missing in the legend. How long do these transitions take?

We added the reaction times to the figure legends.

1. The last part of the discussion states: 'The assembly of an intracellular RED with a membrane-embedded TMD observed in NOX, DUOX, and STEAPs naturally led to the notion that NADPH, FAD, and heme form an uninterrupted rigid electron-transfer chain that shuttles electron from the intracellular cellular NADPH to the extracellular substrates. While this may be true for NOX and DUOX, in which rapid supply of electrons to their extracellular substrates are essential to their biological functions, it may not apply similarly to STEAPs since it has only one heme in the TMD, and their electron transfer relies on shuttling of FAD.' The authors should mention here that the activity of NOX and DUOX is tightly regulated by accessory proteins, Ca²⁺ etc. Similarly, do the authors expect that the large distance between NADPH and FAD in the structures could represent a way to regulate/dampen the metal ion reduction rates of STEAPs in vivo?

We agree. We mentioned the regulation of NOX and DUOX in Discussion. We remain puzzled by the large distance between NADPH and FAD in STEAPs and are in pursuit of a structure in which the two cofactors are “in touch” for electron transfer.